MIL-S-21427A(SHIPS)
21 October 1960
SUPERSEDING
MIL-S-21427(SHIPS)
15 August 1958

MILITARY SPECIFICATION

STRAINER, ASSEMBLIES, MAIN STEAM, HIGH PRESSURE (SIZES 4 INCHES AND ABOVE)

1. SCOPE

1.1 Scope.- This specification covers high pressure steam strainer assemblies in sizes 4 inches and above intended primarily for main steam systems, for protection of power equipment and throttle valves from dirt, sediment, pipe scale and foreign objects.

1.2 Classification.- Strainer bodies shall be of the following compositions as specified (see 6.1):

Composition A - 2-1/4 percent chromium, 1 percent molybdenum alloy.
Composition B - 1-1/4 percent chromium, 1/2 percent molybdenum alloy.
Composition C - Carbon steel.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-D-963 - Drawings, Production, Procedure for Procurement of.
MIL-P-15424 - Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound (for ASA Commercial Flanged Joints in Piping Systems) (Identification Symbol 2410).

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-692-13 - Radiographic Standards for Steel Castings.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN STANDARDS ASSOCIATION
B16.5 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American Standards Association, 70 East 54th Street, New York 17, New York).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS
ASME Boiler and Pressure Vessel Code, Sections I, II and VIII.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 29 West 39th Street, New York, New York).

FSC 4730

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AMERICAN SOCIET. OR TESTING MATERIALS
A-105 - Forged or Rolled Steel.
A-106 - Seamless Carbon Steel Pipe.
A-182 - Forged or Rolled Alloy Steel.
A-216 - Carbon Steel Castings.
A-217 - Alloy Steel Casting.
A-335 - Seamless Ferritic Alloy Steel Pipe.

(Application for copies should be addressed to American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York).

3. REQUIREMENTS

3.1 Material.- Materials shall be as specified in table I. Material for parts other than those listed in table I shall be suitable for the intended pressure and temperature.

Table I - Materials

Name of part	Material form	Composition A	Composition B	Composition C
		Applicable documents		
Body, covers	Forgings	ASTM Publication A-182, F-22	ASTM Publication A-182, F-11	ASTM Publication A-105, grade II
	Castings	ASTM Publication A-217, WC-9	ASTM Publication A-217, WC-6	ASTM Publication A-216, grade WCB
Welded bosses drain connection	Pipe	ASTM Publication A-335, P-22	ASTM Publication A-335, P-11	ASTM Publication A-106
Basket, rims	Plate	Specification QQ-S-766		
Gaskets	Spiral wound	Specification MIL-G-21032, type I		
Nuts, bolts, washers	All material used in the construction of these parts shall be of the quality best suited for the purpose intended and shall be strictly in accordance with the material specification shown on approved drawing.			

3.2 General design.- The strainer assemblies generally shall be composed of a body with inlet connections, closure, strainer basket assembly, bosses for installation of thermometers and either a single or dual outlet as specified (see 6.1). Where strainer body inlet or outlet connections are welded, they shall be limited to 1/2 the size of the strainer body for services 600 pounds per square inch (p.s.i.) and above or 425°F and above.

3.2.1 Pressure temperature ratings.- The design of the strainer bodies shall be in accordance with ASME Boiler and Pressure Vessel Code, Sections I, II and VIII, as applicable. The design pressure shall be equal to 1.05 times maximum system pressure and the design temperature shall be 10°F plus maximum system temperature (see 6.1).

3.2.2 Temperature ratings.- The maximum allowable temperatures for strainers specified herein shall be as follows:

(a) Composition A - 1050°F
(b) Composition B - 1000°F
(c) Composition C - 775°F

3.2.3 Sizes.- The sizes of strainer assemblies, number of inlets and outlets and their sizes, and whether straight, wye, angle or manifold shape shall be as specified (see 6.1).

3.2.4 Strainer baskets.- Strainer baskets shall be made of drilled plate. The welded seam shall not be drilled. The net open area through the basket shall be at least 2 times the inlet pipe area. The plate shall have 3/32 inch maximum drilled openings. The strainer basket shall be strong enough to prevent failure in case it is plugged, a sufficient amount to produce a pressure drop across the basket equivalent to 25 percent of the working pressure. The wall thickness for the strainer basket shall be determined from the formula as specified in Section VIII of ASME Boiler and Pressure Vessel Code. In no case shall the wall thickness be less than 1/4 inch. The ligaments between the perforations shall be determined from the formula as specified in Section I of ASME Boiler and Pressure Vessel Code. The ligament efficiency shall be not less than 30 percent and not more than 50 percent. The holes in the basket shall be so arranged that the maximum continuous unperforated area is in a transverse or circumferential

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direction. The welded seam of the strainer basket shall not be located in the direct line of the main steam flow through the strainer.

3.2.5 Fabrication.- Welding shall be in accordance with Standard MIL-STD-278.

3.2.6 End connections.- Unless otherwise specified (see 6.1), end connections shall be butt welding ends prepared in accordance with ASA Publication B-16.5. For flanged strainer assemblies, flanges in accordance with ASA Publication B-16.5 shall be specified (see 6.1). Flange faces, when flanges are specified, shall be of a finish with a circular lay (concentric or phonographic) having a roughness not exceeding 500 RHR produced by machining cuts of 0.003 inch maximum depth with not less than 40 cuts per inch of face width. For special installations involving radioactive service or hazardous fluids (toxic or explosive) where a finer finish is required, the requirements shall be the same except that the finish shall be 125 RHR maximum. For flanges where the flange face cannot be turned and tool marks run across the flange face, the surface finish shall have a maximum RHR of 125.

3.2.7 End closure.- End closure shall be flanged.

3.2.8 Drain connections.- Each strainer shall be provided with a suitable drain connection. The drain connection shall be so located as to permit adequate drainage of the strainer.

3.2.9 Unless otherwise specified (see 6.1), the strainer shall be designed in such a manner that the steam flow through the strainer basket is from the inside of the basket to the outside of the basket.

3.3 Marking.- All strainer bodies shall have the pressure rating and manufacturer's name or trademark cast or forged integral with the strainer body. No metal stamping shall be permitted on the finished strainer bodies or surfaces subjected to high stresses in service.

3.4 Repair parts.- Flange gasket repair parts shall be furnished 100 percent.

3.5 Drawings.-

3.5.1 Preliminary drawings.- Within 30-days after award of contract, the contractor shall submit to the bureau or agency concerned preliminary drawings that are sufficient to permit an evaluation of the design and approval of materials (see 3.5.1.2). These drawings shall be detail assembly drawings and shall show the following:

(a) A sectional assembly of the strainer.
(b) Bill of material, listing specifications, grade and condition, or other data adequate to identify the materials proposed.
(c) Details of strainer basket.
(d) Dimensions - overall, accessibility space including disassembly clearances and all dimensions pertinent to installation.

3.5.1.2 If a contractor proposes to furnish equipment which previously has been supplied and for which the bureau or agency concerned has approved drawings on record, reference to the approved drawings will be acceptable in lieu of submitting preliminary drawings.

3.5.2 Final drawings.- When specified (see 6.1), detail assembly drawings shall be submitted to the bureau or agency concerned by the contractor for completion of contract or order. These drawings shall be in accordance with Specification MIL-D-963 except for the extent of detail. Only the information required on the preliminary drawings need be furnished. The cognizant inspector shall validate final drawings after receipt of approved preliminary drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Non-destructive inspection.-

4.2.1 Visual and dimensional examination.- Each strainer shall be visually and dimensionally examined to verify conformance with all the requirements of this specification not involving tests.

4.2.2 Pressure containing castings of strainers shall be 100 percent radiographically inspected. Inspection shall be in accordance with Standard MIL-STD-271 and Publication NAVSHIPS 250-692-13. Sections less than 1 inch in thickness shall meet class 1 requirements of Publication NAVSHIPS 250-692-13. Sections with thickness 1 inch and over shall meet class 2 requirements of Publication NAVSHIPS 250-692-13.

4.2.3 Pressure containing forgings of strainers shall be 100 percent magnetic particle inspected in

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accordance with 4.2.2. Forgings shall be free from flaws such as seams, cracks, laps, porosity, scale, flakes, and all other defects detrimentally affecting the suitability of the forgings for the service intended.

4.2.4 Defects to be repaired.- The following shall apply to those parts inspected in accordance with 4.2.3:

(a) Defects less than 0.030 inch in depth need not be repaired provided the bottom of the defect is rounded and visible and the minimum wall thickness is maintained.
(b) Defects greater than 0.030 inch in depth but less than 15 percent of the wall thickness shall be repaired by removing the defective material. This material shall be removed by drilling or grinding to a bottom radius of at least three times the depth of the defect. The depth of the finished repair shall be less than 15 percent of the undamaged wall thickness. All sharp corners shall be faired into the base metal. Welding is not required to effect a repair of this nature, provided the minimum wall thickness is maintained.
(c) All defects greater than 15 percent of the wall thickness shall be repaired by removing the defective material and welding. The material shall be removed by grinding or drilling and then welded in accordance with 3.2.5. The crown of the weld shall be blended into the base metal.

4.2.5 Material inspection.- Material inspection shall be determined as follows:

4.2.5.1 Pressure containing parts shall be inspected to determine conformance with the applicable material specification.

4.2.5.2 The inspector shall normally accept certification that nonpressure containing parts conform to their applicable material specification providing that such certification contains actual test, inspection, or other verifiable quality data.

4.2.5.3 When there is a question relative to material identification, the inspector may make a chemical analysis and mechanical tests as he deems necessary to establish complete identity and overall quality.

4.3 Tests.-

4.3.1 Hydrostatic tests.- Each of the strainers shall be subjected to the tests specified in 4.3.1.1 for strength and porosity. The water temperature shall not exceed 100°F.

4.3.1.1 Shell test.- Strainer shall be subjected to pressures determined by the following formula for a duration of 10 minutes (min.). Any weeping, leakage, or permanent deformation shall be cause for rejection:

$$P_h = 1.5 \left(P_d \times \frac{S_r}{S_o}\right)$$

Where:

P_h = Required hydrostatic pressure (p.s.i.g.)
P_d = Design pressure of the strainer (p.s.i.g.)

S_o = Allowable stress at design temperature (p.s.i.)
S_r = Allowable stress at room temperature (p.s.i.)

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.-

5.1.1 Strainer assemblies.- Unless otherwise specified (see 6.1), strainer assemblies shall be preserved and packaged in accordance with the manufacturer's commercial practice. Packing shall be in a manner which will insure acceptance by common carrier and safe delivery at destination. Shipping containers or method of packing shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.2 Repair parts.- Repair parts shall be preserved, packaged and packed for the level specified (see 6.1), in accordance with Specification MIL-R-196 and MIL-P-15424, as applicable. Repair parts not covered therein shall be processed in accordance with the guide lines of Specification MIL-P-116. Repair parts shall be individually packaged unless used in sets or quantities greater than one.

5.2 Marking.- In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition required (see 1.2).
(c) When either a single or dual outlet is required (see 3.2).
(d) Design pressure and design temperature (see 3.2.1).

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(e) Maximum allowable temperatures (see 3.2.2).
(f) Sizes of strainer assemblies, number of inlets and outlets, and shape (see 3.2.3).
(g) End connections and flanges required (see 3.2.6).
(h) Direction of steam flow (see 3.2.9).
(i) When final drawings are required (see 3.5.2).
(j) Preservation, packaging and packing required for strainer assemblies (see 5.1.1).
(k) Preservation, packaging and packing required for repair parts (see 5.1.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy-Bureau of Ships
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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